Instructions for the Independent Project

Molecular Evolution (Spring 2026)

For your Independent Course Project, you will be assigned a gene and a focal species. Details on your gene and focal species can be found in a separate document on Studium. You will investigate the gene and the dN/dS values along the tree of the focal species in order to discuss the potential role of natural selection on the evolution of this gene. For that, you should follow the procedure that we use in Lab 2. Keep in mind that it is important to go through the entire protocol of Lab 2 to characterise the selection regime in this gene.

You will need to download the coding sequence of the gene in your focal species as well as at least 6 - 7 other species. You should search for the gene on the Ensembl homepage, selecting the species you were given (you may need to select Ensembl Plants from the bottom of the homepage). You can then view orthologues of this gene in other species from the list. You should choose species that have **1-to-1 orthologues** with the species you were assigned. You can also select the gene tree option in the menu on the left hand side which will show you the evolutionary relationships, and allow you to choose orthologous sequences that are an appropriate evolutionary distance. As a guideline, you can begin by looking for species within the same family/order as your focal species, however you may need to look at greater distances than this. You can also use the branch lengths in the gene tree you estimate to help guide you. A rule of thumb is to **avoid branch lengths shorter than 0.01** (branch lengths can be seen on the gene-tree by clicking on the branch). You will be introduced to searching Ensembl at the start of Lab 2.

After you decide what species to include, you will follow the steps from Lab 2 to download your sequences, build the tree file, and test selection patterns along the tree.

Project Report

Once you have finished your analysis, you will write a report for this project. You should include an Introduction, Methods, Results, Discussion, and References section in your report. To write the Introduction, you should search for literature on the gene you were assigned and discuss what is known already. What is the function of this gene? Is there anything known already about selection on this gene either in your own focal species or other species? When you discuss this background you should include references within the text of your report, which are to be listed in a References section at the end of the report. You have been given a short demonstration on how to cite references during the compulsory lecture introducing the Independent Course Project (see recorded lecture in Studium). You should also describe the methods that you used, and present a summary of your results. Finally, you should discuss what your results mean in the context of selection on this gene.

In addition to submitting your Project Report, we would like you to submit the code used to perform the analyses. This means a documentation or script files of how you ran your sequence alignment, and what you specified in your codeml control files. This can be added in the main report as supplemental material towards the end of the report.

Length of the Project Report

The length of the complete project report should be no more than four A4 pages. Some other specifics pertaining to the length:

- Single-spaced
- References section not included in the 4-page limit
- Code/script not included in the 4-page limit

The report will be submitted through Studium.

Grading criteria

For the independent project you will be graded according to the following criteria.

Fail: did not fulfill the requirements of the report OR if it is clear that the report is AI generated.

Pass: Descriptive description of the material, identifies key concepts and applies correctly to data

Pass + 2 points: Critically evaluate concepts, acknowledging limitations and strong narrative flow

Pass + 4 points: Demonstrate systematic integration of diverse scientific sources (original literature, lectures etc) to build a high level reflective and scientifically accurate reflection.

Deadline for Submission of the Project Report

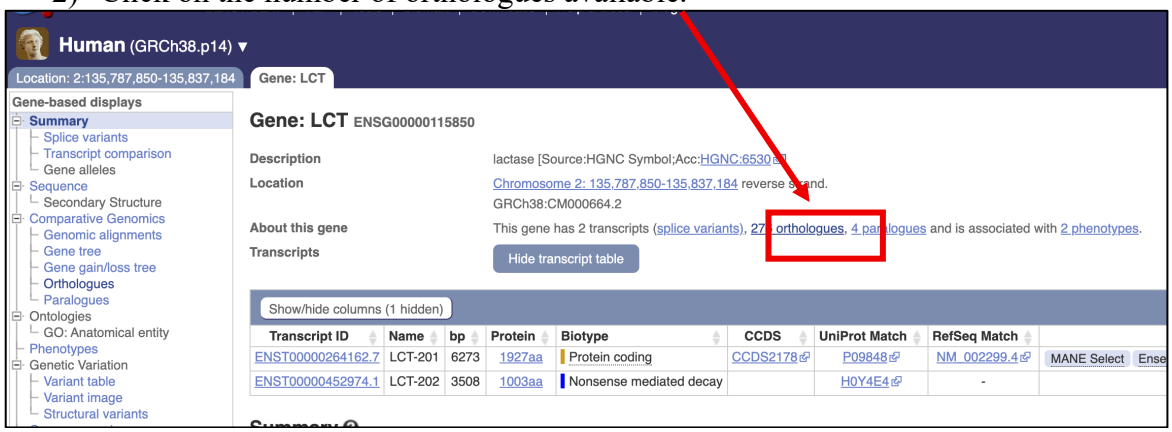
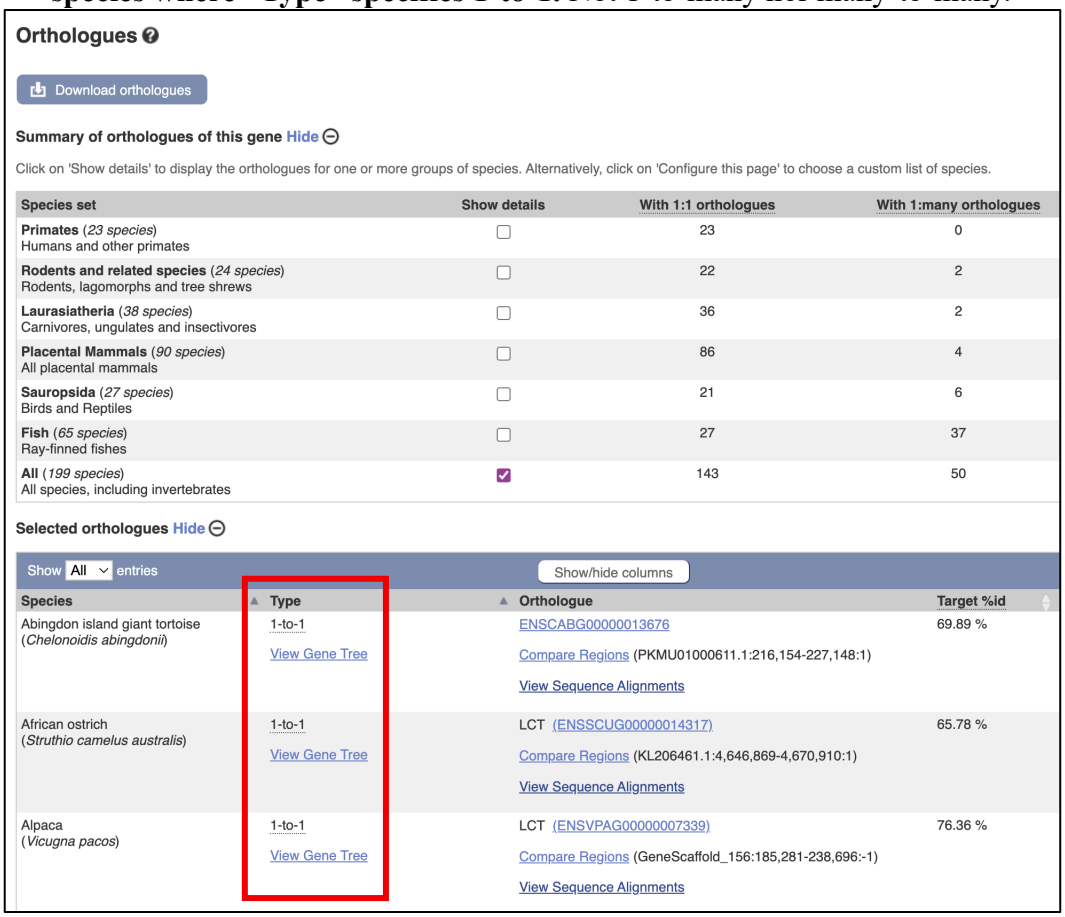
Deadline: 3rd March

How to choose 1:1 orthologous sequences on Ensembl

It is important to select species which have a 1:1 orthologue with the gene + species combo you have been assigned. In order to do so you can follow this simple guide:

Assuming I have been assigned the LCT gene and the Human as a focal species:

1) Navigate to the gene page of your species on Ensembl.
2) Click on the number of orthologues available.



3) You will find the following Orthologue table below with a list of species. **Choose species where "Type" specifies 1-to-1.** Not 1-to-many nor many-to-many.